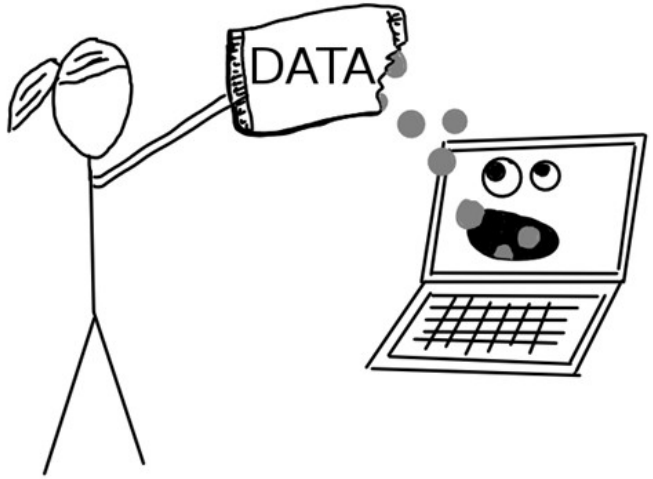
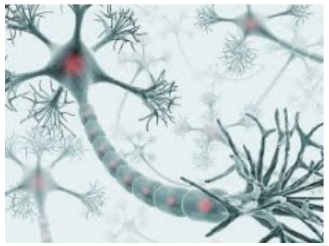
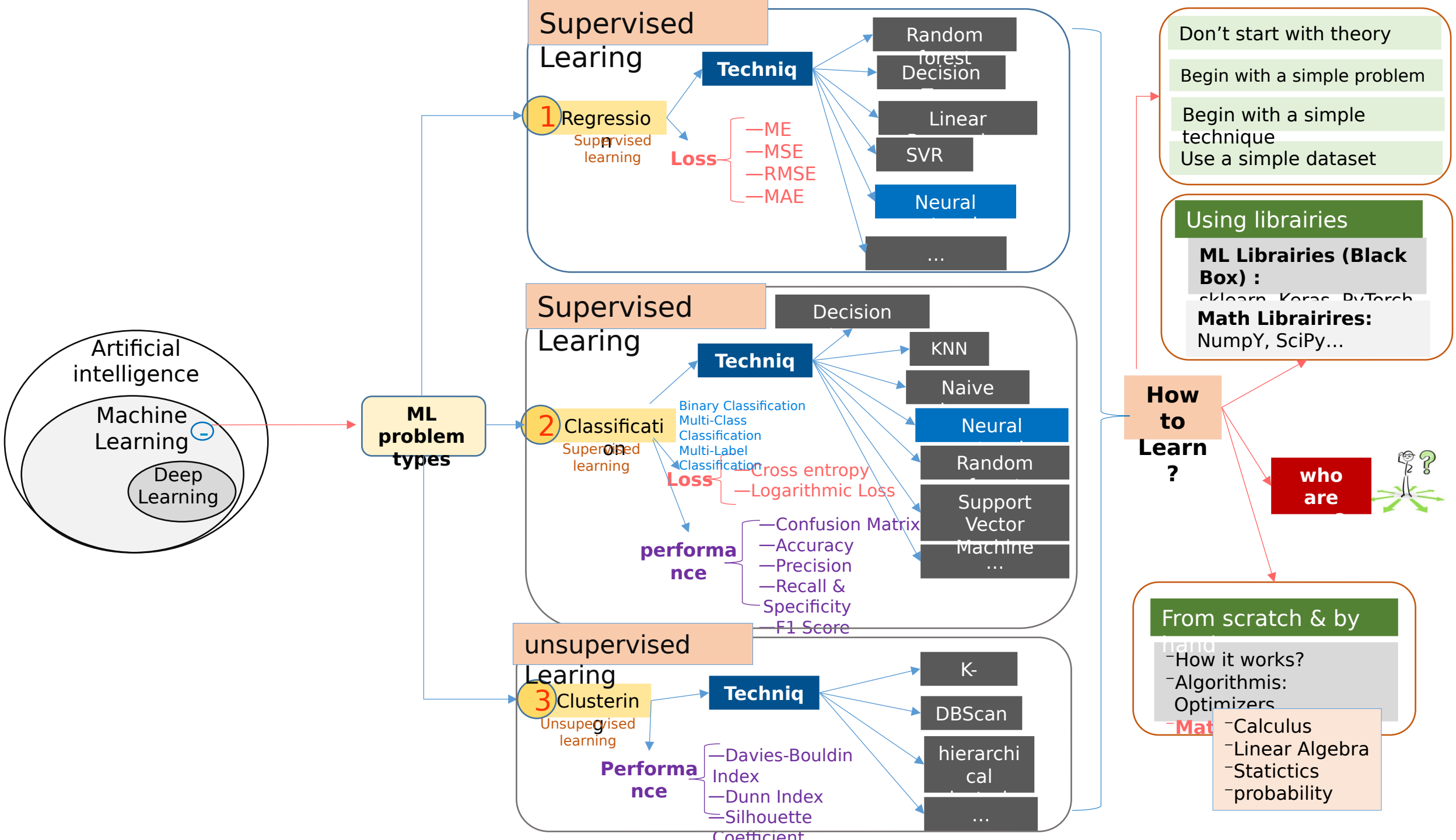




Linear Regression
Logistic Regression
Neural Network

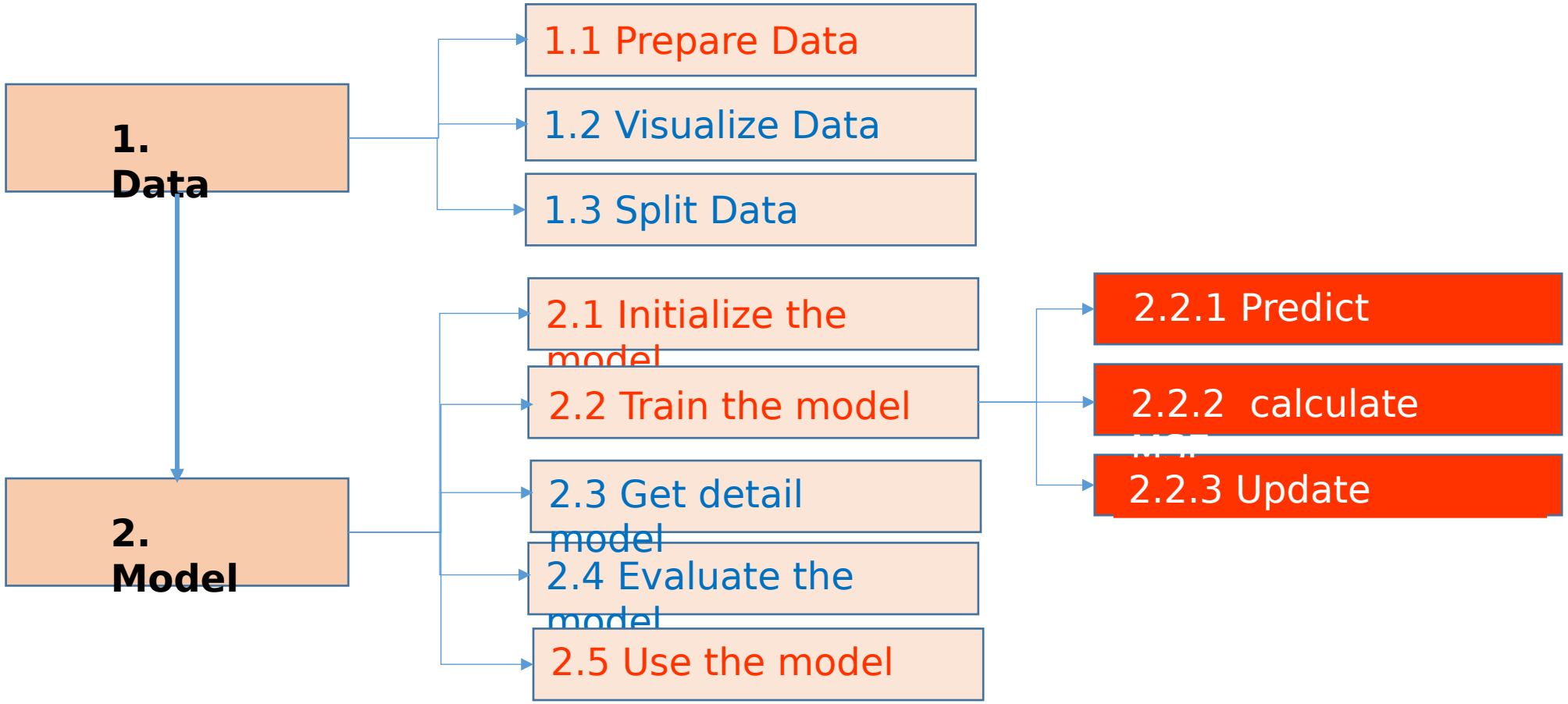
2022/20

Abdelwahab Naji - Enseignant chercheur
Abdelwahab.naji@gmail.com
ENSAO, Université Mohamed premier - Oujda
https://www.youtube.com/c/AbdelwahabNaji


Artificial intelligence
Machine Learning
Deep Learning
ML problem types
Supervised Learing
1 Regression
Supervised learning
Techniq
Loss
—ME
—MSE
—RMSE
—MAE
Random forest
Decision
Linear
SVR
Neural
...
Supervised Learing
2 Classification
Supervised learning
Techniq
Binary Classification
Multi-Class Classification
Multi-Label Classification
Loss
—Cross entropy
—Logarithmic Loss
performance
—Confusion Matrix
—Accuracy
—Precision
—Recall & Specificity
—F1 Score
Decision
KNN
Naive
Neural
Random
Support Vector Machine
...
unsupervised Learing
3 Clustering
Unsupervised learning
Techniq
Performance
—Davies-Bouldin Index
—Dunn Index
—Silhouette Coefficient
K-
DBScan
hierarchical
...
How to Learn ?
Don't start with theory
Begin with a simple problem
Begin with a simple technique
Use a simple dataset
Using librairies
ML Librairies (Black Box) :
sklearn, Keras, PyTorch
Math Librairires:
NumpY, SciPy...
who are
From scratch & by hand
-How it works?
-Algorithmis: Optimizers
-Mat
-Calculus
-Linear Algebra
-Statictics
-probability

Machine Learning

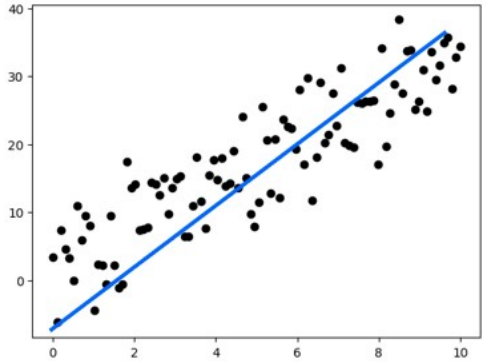
Les éléments + étapes



Neural Network for Supervised Learning

1

Simple Linear Regression Problem



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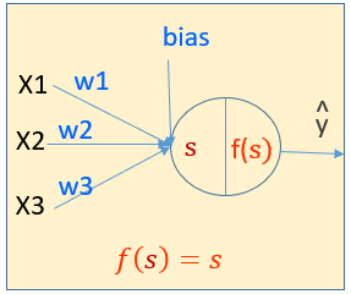
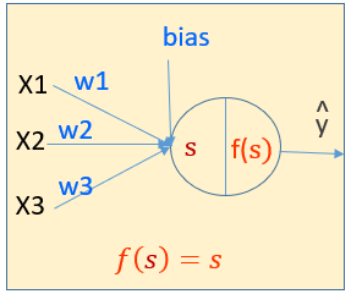


2

Multi Variate Linear Regression Problem

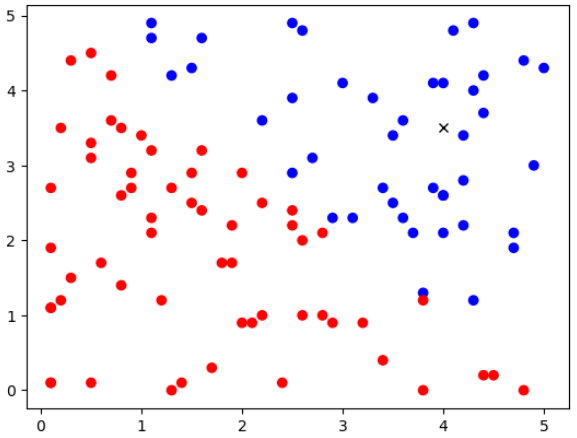
TV	Radio	Newspaper	CA
230.1	37.8	69.2	**22,1**
44.5	39.3	45.1	**10,4**
17.2	45.9	69.3	**9,3**
...			
199.8	2.6	21.2	**10,6**

Advertising p

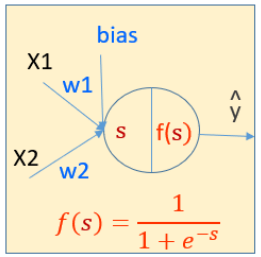


3

Linearly Separable problem

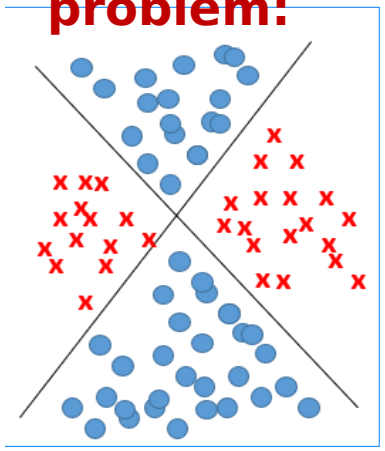


Red or Blue?

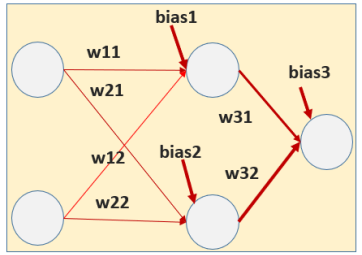


4

Non Linearly Separable problem:

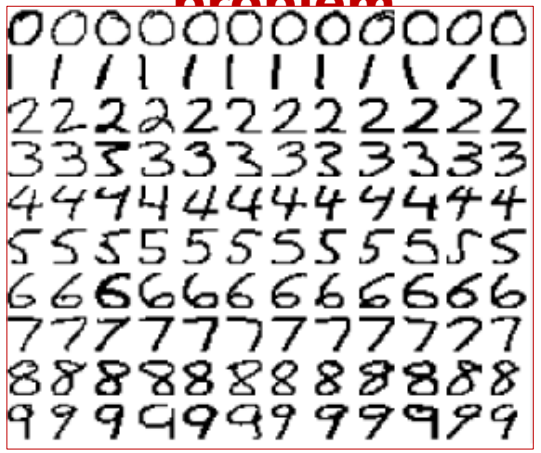


XOR Problem

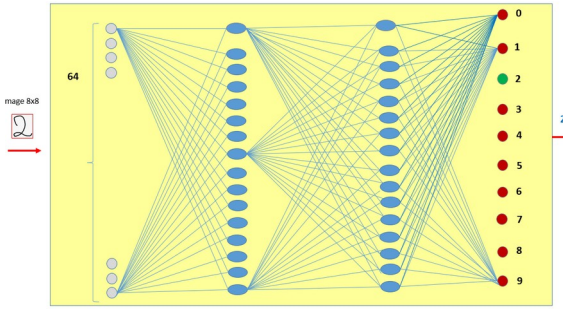


5

Complex classification problem



Hand Written digits



Exemple

- Une société souhaite analyser le chiffre réalisé (**Sales**) en fonction du budget alloué à la publicité (frais)
- La publicité se fait en utilisant différents plateformes média (TV, Radio, newspaper)
- **Objectif:** prédire le total des ventes (sales) selon les frais de publicité

Data Set

TV	Radio	Np	CA
230.1	37.8	69.2	**22,1**
44.5	39.3	45.1	**10,4**
17.2	45.9	69.3	**9,3**
151.5	41.3	58.5	**18,5**
180.8	10.8	58.4	**12,9**
8.7	48.9	75	**7,2**
57.5	32.8	23.5	**11,8**
120.2	19.6	11.6	**13,2**
8.6	2.1	1	**4,8**
199.8	2.6	21.2	**10,6**

X1=[230.1,37.8,69.2]

y1=22.1

Sales à prédire

TV	Radio	Newspaper	**Sales** (target)
66.1	5.8	24.2	?
214.7	24	4	?
23.8	35.1	65.9	?

- Dans le Dataset chaque exemple d'apprentissage est accompagné par une target: Supervised Learning (Apprentissage supervisé)
- Objectif: Prédire une valeur continue
problème de régression

- (X_i, y_i): un exemple d'apprentissage
- features: Tv, Radio, newsPaper (X)
- target: sales (y)

Sales_predicted = bias+w_1*X1+W_2*X2+ W_3*X3

https://www.kaggle.com/ashydv/advertising-dataset

Si les valeurs de X sont grandes, afin d'éviter des mises à jours des poids très grandes, il faut standardiser les données

```
from sklearn.preprocessing import StandardScaler
scaler= StandardScaler()
X_st = scaler.fit_transform(X)
```

$$z = \frac{x - \mu}{\sigma}$$

- z: variable standardisée
- μ: Moyenne
- σ: Déviation standard

1 feature, le modèle est: bias+w1*X1

```
lr=0.01
def fit(X,y):
    n=len(y)
    w1=0
    bias=0
    for k in range(1000):
        yhat=bias+w
        dbias=-(2/n)*np.sum(y-yhat)
        dw1=-(2/n)*np.sum(X*(y-yhat))
        bias=bias-dbias*lr
        w1=w1-dw1*lr
```

initialize the model

Predict using the model

Update the model

```
def predict(X1):

    return bias+w1*X1

#penser à la remise à l'échelle si X1 est standardisées
```

3 features, le modèle est: bias+w1*X1 +w2*X2 +w3*X3

```
lr=0.002
def fit(X1,X2,X3,y):
    n=len(y)
    w1=0
    w2=0
    w3=0
    bias=0
    for k in range(10000):
        yhat=bias+w1*X1+w2*X2+w3*X3

        dbias=-(2/n)*np.sum(y-yhat)

        dw1=-(2/n)*np.sum(X1*(y-yhat))
        dw2=-(2/n)*np.sum(X2*(y-yhat))
        dw3=-(2/n)*np.sum(X3*(y-yhat))

        bias=bias-dbias*lr

        w1=w1-dw1*
        w2=w2-dw2*
        w3=w3-dw3*
    return bias,w1,w2,w3
```

initialize the model

Predict using the model

Update the model

X1, X2 et X3 doivent être standardisés

```
def predict(X1,X2,X3):

    return
bias+w1*X1+w2*X2+w3*X3

#penser à la remise à l'échelle si X1,X2,X3 sont standardisées
```

m features, le modèle est: bias+w1*X1 +w2*X2 +w3*X3+… +wm*Xm

X a m features
Et si on veut que fit soit capable de traiter m features!!!

```
lr=0.001
def fit(X,y):
    n=len(y)
    w1=0
    w2=0
    w3=0
    bias=0
    for k in range(10000):
        yhat=bias+w1*X1+w2*X2+w3*X3
        dbias=-(2/n)*np.sum(y-yhat)

        dw1=-(2/n)*np.sum(X1*(y-yhat))
        dw2=-(2/n)*np.sum(X2*(y-yhat))
        dw3=-(2/n)*np.sum(X3*(y-yhat))

        bias=bias-dbias*lr

        w1=w1-dw1*lr
        w2=w2-dw2*lr
        w3=w3-dw3*lr
    return bias,w1,w2,w3
```

Initialize the model

```
W=np.array([0,
W=np.array(np.zeros(X.shape[1]))
```

Predict using the model

```
yhat=bias+np.dot(X,W)
```

Update the model

```
dW=-(2/n)* np.dot(X.T,(y-yhat))
W=W-dW*lr
```

```
lr=0.001
def fit(X,y):
    n=len(y)
    #W=np.array([0,0,0])
    W=np.array(np.zeros(X.sha
     # X.shape[1] est le nombre d'attributs
    bias=0
    for k in range(10000):
        yhat=bias+np.dot(X,W)

         dbias=-(2/n)*np.sum(y-
         dW=-(2/n)* np.dot(X.T,(y-yhat))
        bias=bias-dbias*lr
         W=W-dW*lr

    return bias,W
```

initialize the model

Predict using the model

Update the model

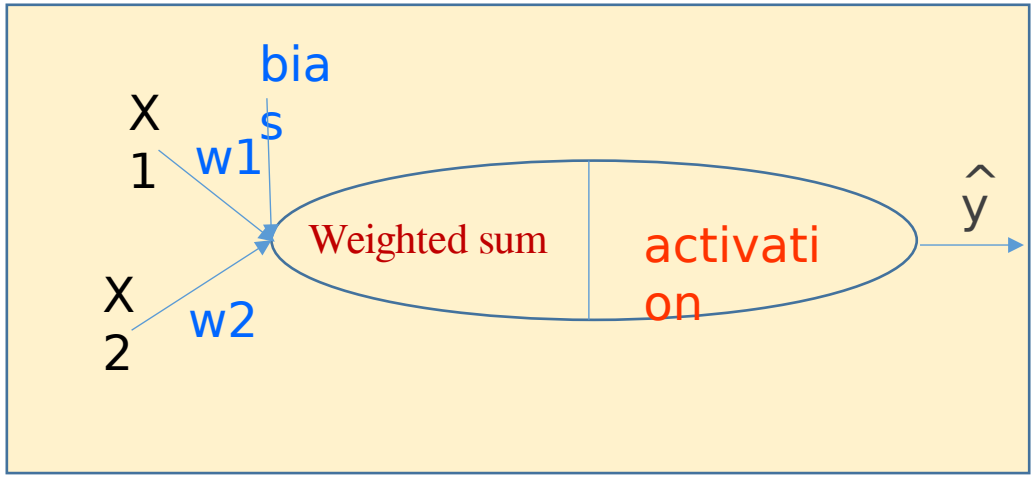
```
def predict(X):

    return
bias+np.dot(X,W)
```

Si X1, X2 et X3 ont été standardisés,

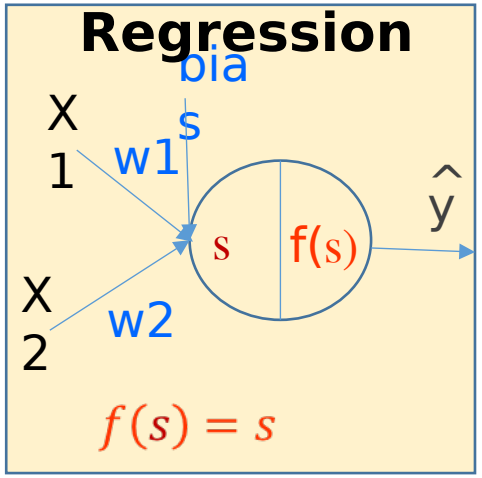
il faut les rendre à leurs échelles

Réutiliser le même code source pour créer un réseau de neurone (un seule neurone)



Regression

Linear Regression



$$weighted\ sum = bias + w1 * X1 + w2 * X2$$

Activation: une fonction d'activation

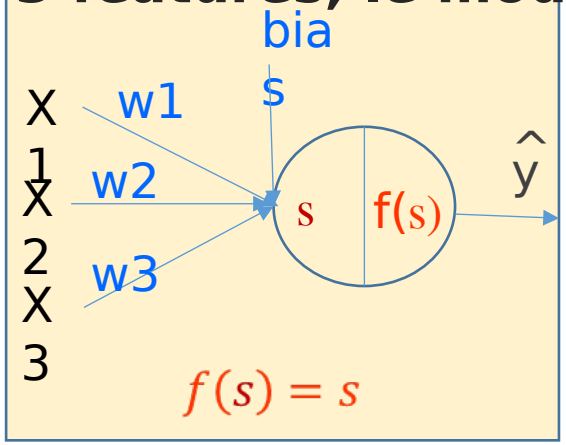
- **Identity or linear**
- Sigmoid
- Tanh
- Relu
- Leaky ReLU
- …

Le choix de la fonction d'activation dépend de…

Code source

Regression Problem by using neural network (1 neurone)

3 features, le modèle est:



$weighted\ sum = bias + w1 * X1 + w2 * X2 + w3 * X3$

Activation: Identity or linear

```
def identity(z):
    return z
```

```
def predict(X1,X2,X3):

ws=bias+w1*X1+w2*X2+
w3*X3

    yhat=identity(ws)
```

```
lr=0.001
def fit(X1,X2,X3,y):
    n=len(y)
    w1=0
    w2=0
    w3=0
    bias=0
    for k in range(10000):
        ws=bias+w1*X1+w2*X2
        yhat=identity(ws)

        dbias=-(2/n)*np.sum(y-yhat)

        dw1=-(2/n)*np.sum(X1*(y-yhat))
        dw2=-(2/n)*np.sum(X2*(y-yhat))
        dw3=-(2/n)*np.sum(X3*(y-yhat))

        bias=bias-dbias*lr

        w1=w1-dw1*lr
        w2=w2-dw2*lr
        w3=w3-dw3*lr
    return bias,w1,w2,w3
```

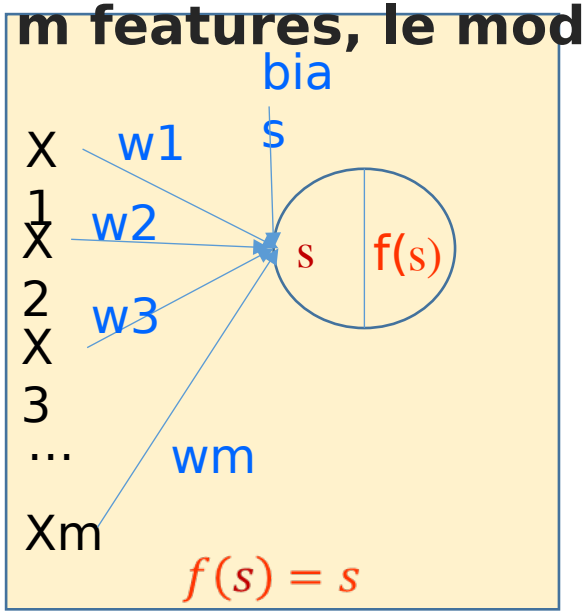
initialize the model

Predict using the model

Update the model

Regression Problem by using neural network (1 neurone)

m features, le modèle est:



$$weighted\ sum = bias + w1 * X1 + \cdots + wm * Xm$$

Activation: Identity or linear

```
def identity(z):
    return z
```

```
def predict(X):
    ws=bias+np.dot(X,W)
    yhat=identity(ws)
    return yhat
```

```
lr=0.001
def fit(X,y):
    n=len(y)
    #W=np.array([0,0,0])
    W=np.array(np.zeros(X.sha
     # X.shape[1] est le nombre d'attributs
    bias=0
    for k in range(10000):
        ws=bias+np.dot(X,W)
        yhat=identity(ws)

        dbias=-(2/n)*np.sum(y-yhat)
        dW=-(2/n)* np.dot(X.T,(y-yhat))
        bias=bias-dbias*lr
        W=W-dW*lr

    return bias,W
```

initialize the model

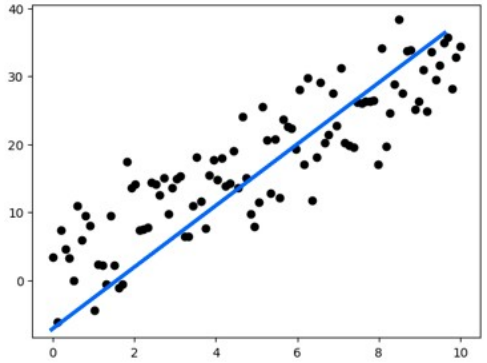
Predict using the model

Update the model

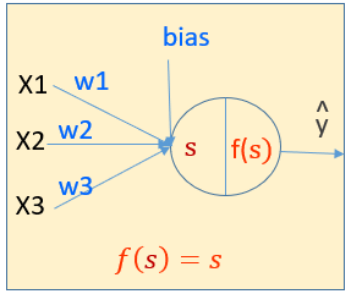
Neural Network for Supervised Learning

1

Simple Linear Regression Problem



Random point cloud



2

Multi Variate Linear Regression Problem

TV	Radio	Newspaper	CA
230.1	37.8	69.2	**22,1**
44.5	39.3	45.1	**10,4**
17.2	45.9	69.3	**9,3**
...			
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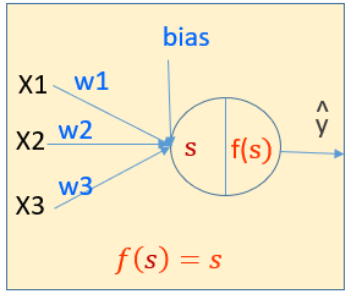
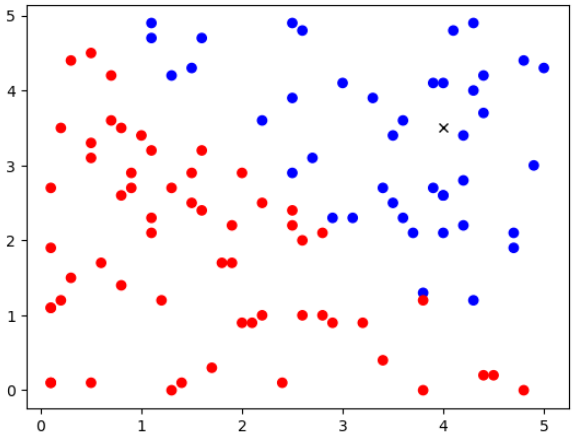
Advertising p

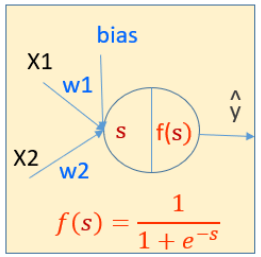


3

Linearly Separable problem

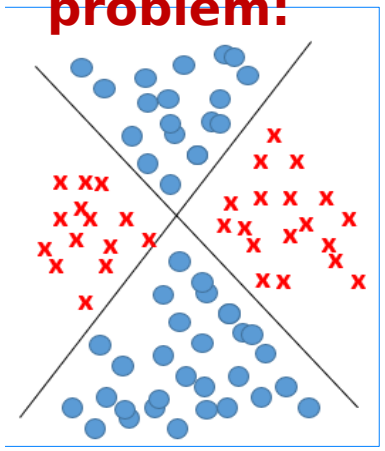


Red or Blue?

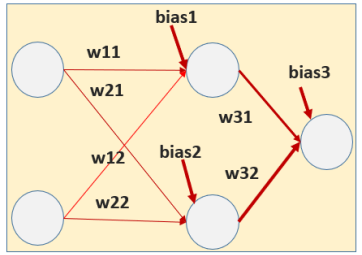


4

Non Linearly Separable problem:

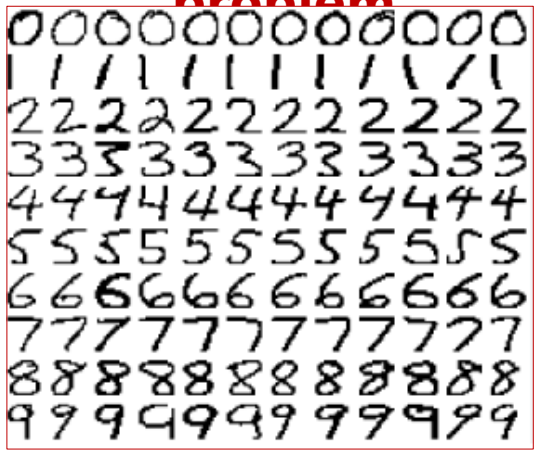


XOR Problem

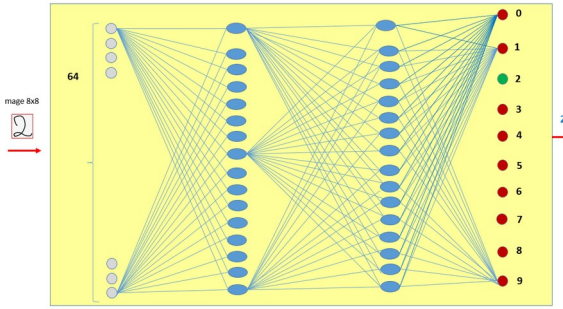


5

Complex classification problem



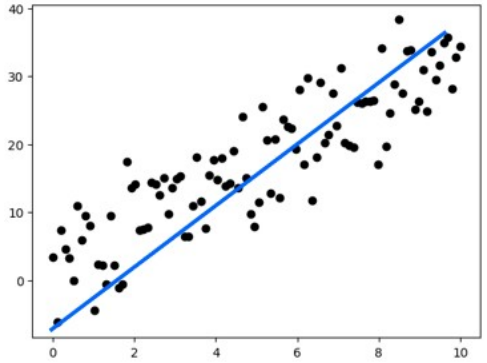
Hand Written digits



Neural Network for Supervised Learning

1

Simple Linear Regression Problem



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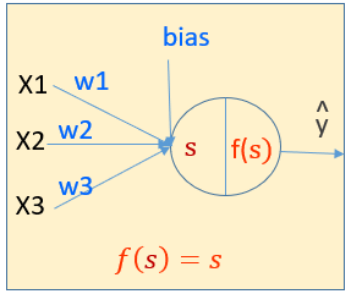

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Multi Variate Linear Regression Problem

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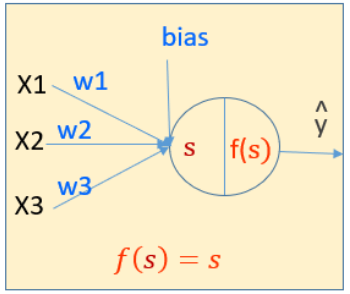
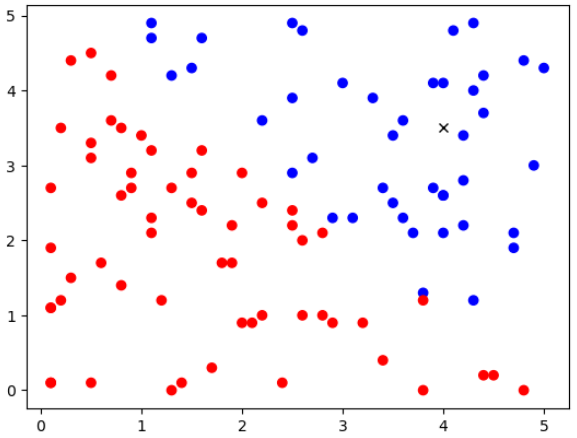
...			
199.8	2.6	21.2	**10,6**

Advertising p

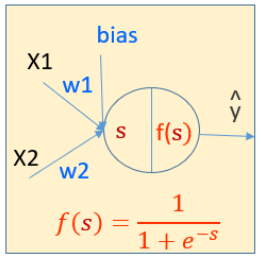


3

Linearly Separable problem

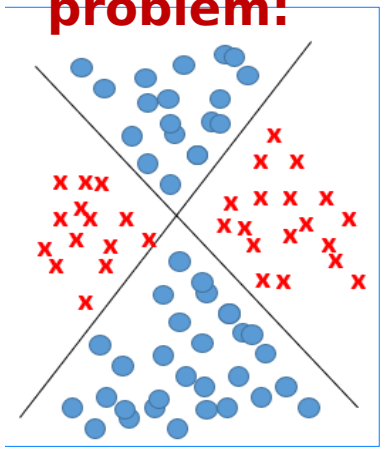


Red or Blue?

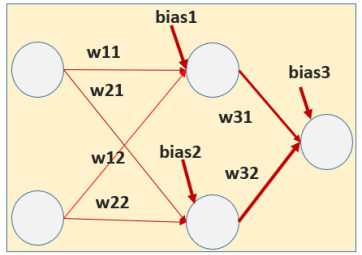


4

Non Linearly Separable problem:

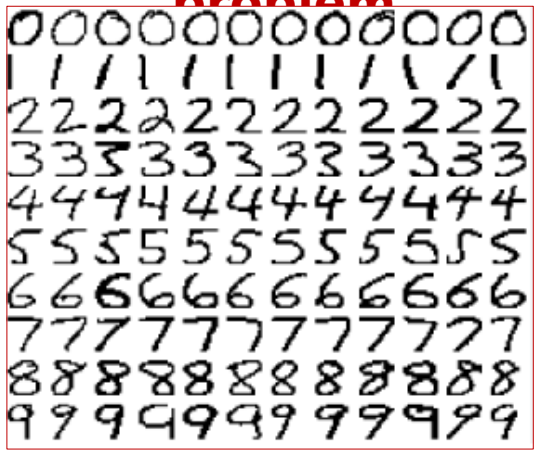
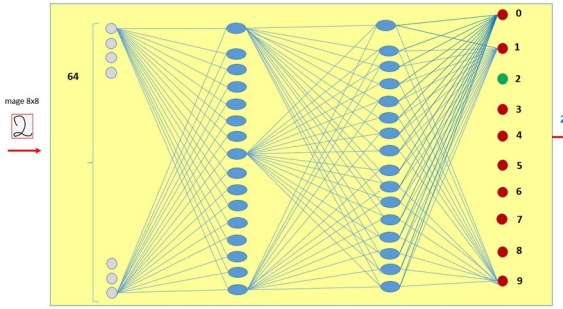


XOR Problem



5

Complex classification problem



Hand Written digits